Supplementary material

This supplement accompanies the revised manuscript on region selection and acquisition order for CAI assessment. All numerical results, intervals, trajectories and diagnostic arrays were already saved. This revision performs deterministic source binding, formatting and plotting only. No optimizer update, research-model forward, new bootstrap draw, seed, reserved-partition evaluation or checkpoint selection was performed.

S1. Data identity, partitions and evaluation units

The six paired-image collections and strength labels are the existing references [1,2,9–15] in the main article. A specimen record binds its source ID, specimen ID, registered image pair and CAI label. Capture groups were formed before splitting from shared source/crop identities, then retained within partitions. The split contains 276 physical specimens (161 training, 50 development validation and 65 reserved), with 152/48/59 capture groups. The reserved partition was not evaluated. The same 50 development specimens selected checkpoints and estimated effects; all intervals are conditional on that selection.

Table S1. The complete 50-specimen evaluation cohort. IDs are source-qualified; dimensions and area clocks remain in cohort_manifest.csv. The full 276-row partition metadata are provided as S1_all_partition_metadata.csv. Reading split labels is not reserved-set evaluation.

Specimen	Capture group	CAI (MPa)
74t7kcdgkr:c8-12	cg_306e4a499acc46c1	197.97145080566406
74t7kcdgkr:c8-13	cg_f6941b7d17a6b65f	266.5553894042969
74t7kcdgkr:c8-16	cg_f6941b7d17a6b65f	264.1634521484375
74t7kcdgkr:c8-2	cg_8ffe4fd6a878d7f3	163.86158752441406
74t7kcdgkr:c8-21	cg_49d6f1237b716b24	263.10400390625
74t7kcdgkr:c8-28	cg_a983245f10699099	282.8260803222656
74t7kcdgkr:c8-3	cg_8ffe4fd6a878d7f3	213.09927368164062
74t7kcdgkr:c8-39	cg_ff43edde8af2ccdb	168.51828002929688
74t7kcdgkr:c8-43	cg_a37dc8a00cc334eb	199.83413696289062
cgtnjyggtm:q24-11	cg_02fd3fc8caefe377	300.3553771972656
cgtnjyggtm:q24-22	cg_54890e18ff2b2669	415.8692932128906
cgtnjyggtm:q24-24	cg_39bf4ddc679c9a42	428.6219482421875
cgtnjyggtm:q24-30	cg_d3f7feef05d5a66a	283.3366394042969
cgtnjyggtm:q24-44	cg_9ba3cd5fb00ea91f	109.75041198730469
cgtnjyggtm:q24-47	cg_523711040788bde2	119.86738586425781
cgtnjyggtm:q24-48	cg_5549ad4618fa66a9	136.6800079345703
cgtnjyggtm:q24-6	cg_344fc945dcb88757	348.28472900390625
cgtnjyggtm:q24-9	cg_2cc9cac1c5baaf14	262.2107238769531
w68dtmpfyf:q16-14	cg_73ef27ab04a735de	404.2454528808594
w68dtmpfyf:q16-2	cg_ff2d01cab2de3ddc	304.9983215332031
w68dtmpfyf:q16-29	cg_94a031a5fc453cba	338.54833984375
w68dtmpfyf:q16-3	cg_3361913654df8b27	352.2940673828125
w68dtmpfyf:q16-33	cg_a37eab4f5b7beb12	257.2702331542969
w68dtmpfyf:q16-38	cg_a052ba6f5200ee4f	344.6358337402344
w68dtmpfyf:q16-43	cg_6853c8e74a68016b	302.281982421875
w68dtmpfyf:q16-6	cg_ce4d6a48a6968060	418.267333984375
xcmzfsbd9t:c24-37	cg_a01963fde121eda0	381.1910705566406
xcmzfsbd9t:c24-39	cg_d72ecf3171227da2	246.54722595214844
xcmzfsbd9t:c24-42	cg_a5c5defb812088ae	301.55389404296875
xcmzfsbd9t:c24-47	cg_954e3ed61c8db0a4	480.5938415527344
xcmzfsbd9t:c24-50	cg_59e95410b683945f	292.1376647949219

Specimen	Capture group	CAI (MPa)
xcmzfsbd9t:c24-51	cg_7e1c8f7865d73e78	263.2923583984375
xcmzfsbd9t:c24-52	cg_b98b3f9ca1bfb997	252.29762268066406
xcmzfsbd9t:c24-56	cg_06ec34425b30c2e6	230.2566680908203
xcmzfsbd9t:c24-9t	cg_0d1994d0164384d4	278.2536315917969
yfxyg8jm46:c16-12	cg_5a80e5179694ff19	268.03814697265625
yfxyg8jm46:c16-14	cg_efd8928c6d17de81	508.2288818359375
yfxyg8jm46:c16-16	cg_cd055af98e2ea119	499.259765625
yfxyg8jm46:c16-17	cg_f64812b6759e7d2b	512.5420532226562
yfxyg8jm46:c16-24	cg_354aea2e84731e86	523.44140625
yfxyg8jm46:c16-39	cg_514ca9d914207c05	234.76339721679688
yfxyg8jm46:c16-42	cg_cb9c6f640f08aee7	322.6684265136719
yfxyg8jm46:c16-5	cg_7f9fe4a18a151cd4	270.7127685546875
ykhs7s2dck:q8-15	cg_60c6812942ba2332	260.08502197265625
ykhs7s2dck:q8-2	cg_93e604617899fa5f	219.15264892578125
ykhs7s2dck:q8-22	cg_ca08b9786e924e66	235.01319885253906
ykhs7s2dck:q8-33	cg_a841812423416444	215.7657928466797
ykhs7s2dck:q8-34	cg_f28b9720dbc7e026	235.4805145263672
ykhs7s2dck:q8-37	cg_a064df3d047f4519	248.33407592773438
ykhs7s2dck:q8-8	cg_8fd87d72a8d71b83	184.39810180664062

Random contributes five saved trajectories per specimen; the other five methods contribute one each. Thus 500 archived trajectories and 7,849 acquisition events are repeated observations of these 50 specimens, not additional independent samples. Fixed-final-set analysis adds seven conditions per specimen (original, reverse, five permutations), giving 350 conditions on the same cohort. Paired intervals resample capture groups within six source groups, carrying each group's specimens together.

S2. Model identity, representation and learning protocol

The frozen actor is SpatialCAIActor with use_vlm=False and use_feedback=True. The method key is `NO_VLM_SPATIAL_FEEDBACK`; its seed is 2026091302, selected update 1000 after 1250 completed updates. The frozen readout is MeanSCPredictor (MEAN_SC), selected at update 1750 after 2000 updates. The actor contains 378,978 parameters including its training value head. The model-identity lock records the checkpoint and state-dictionary identities and whether local checkpoint bytes could be rehashed. A manifest-only checkpoint verification is not represented as a new model load.

Table S2. Frozen architecture and observable-input contract.

Component	Specification
Local encoder	Frozen ResNet18; independently crop each cell before encoding; 512 features per region
Grid / native area	8 × 8 row-major, indices 0–63; edges rint(linspace(0, native dimension, 9)); integer pixels
Surface registration	Frozen orientation; clockwise 90° once; maximum edge 1024 before local extraction
Actor local projections	Surface 512 → 48; visible/learned-unknown internal 512 → 48
Actor regional input (102)	48 surface + 48 internal + 2 coordinates + 1 measured + 1 ordinal/64 + 2 zero VLM placeholders

Component	Specification
Actor global input (6)	Acquired area, remaining area, standardized CAI, measured-cell fraction, 2 zero VLM placeholders
Actor contextualizer	64 regional + 1 global tokens; width 128; 2 layers; 4 heads; feed-forward 256; dropout 0
Actor heads	Concatenate local/global tokens → 128 → 1 score; separate global value head during training
Readout local cell	Two 512 → 64 affine projections; concatenate with mask and coordinates (131 inputs); 131 → 64 → 64 GELU
Readout pooling / output	Mean of all 64 cells + mean of measured cells + acquired area (129 inputs); 129 → 64 → 1; training-only target scaling
Readout missing input	Internal descriptor is zero-masked BEFORE affine projection, retaining bias; measured-cell mean is zero at empty mask
Readout order dependence	Set-based; no ordinal or history input. Same final set and area imply the same terminal input.

The actor uses a learned unknown internal descriptor and an explicit measurement indicator. It never treats an unmeasured internal cell as known image evidence. Its two reserved regional channels and two reserved global channels are constant zero in this model. The predictor instead masks raw internal descriptors before its affine projection, so an unmeasured cell still contributes the projection bias to the all-cell summary. These are different missing-input mechanisms. Surface descriptors are available initially; measured CAI enters training costs or retrospective metrics only, never an execution decision.

Predictor mask training uses the frozen mixture: 10% empty masks; 60% masks with 1–16 cells drawn from random or fixed geometric orders; 20% random masks with 17–48 cells; 10% full masks. The standardized residual is fitted with Huber loss. Three capture-group cross-fitted training predictors provide policy cost: a training specimen is scored only by the predictor excluding its fold. The common evaluation predictor is fitted on the training partition. This description documents the completed training; the revision does not execute it.

Policy optimization used AdamW with learning rate 3×10^{-4}, weight decay 10^{-4}, batch size 16 and gradient-norm limit 1. Training sampled feasible categorical actions. Evaluation used deterministic feasible-score argmax. Candidate checkpoints were compared every 250 updates on development process A; an earlier candidate was replaced only for an improvement exceeding 10^{-12}. The following scores are the existing job history, not a new candidate comparison.

Table S3. Archived actor-selection history.

Update	Development A (MPa)
250	47.276
500	46.461
750	44.911
1000	43.597
1250	44.451

S2.1. Derivation and cost-minimization sign

All 13 original numbered display equations remain in the main article; equation_map.csv records their locations. To derive Eqs. (9–10), substitute the signed error identity below into the interval sum in Eq. (7). Each completed

event contributes to every later interval, with total length B − c_j including final carry. Unequal cell areas and negative decreases are retained.

$$e_t = e_0 - \sum_{j=1}^{t} \Delta e_j, \qquad \sum_{k=j}^{T-1} (c_{k+1} - c_k) + (B - c_T) = B - c_j.$$

$$A(B) = e_0 - \sum_{j=1}^{T} (1 - c_j/B)\Delta e_j, \qquad J = (1 + \lambda_T)e_0 - \sum_{j=1}^{T} (1 - c_j/B + \lambda_T)\Delta e_j.$$

The trajectory baseline e_0 is constant for a fixed specimen and predictor. An earlier favorable event has more remaining integration length, whereas an adverse event contributes negatively. At c_j = B the process contribution vanishes and the terminal weight $\lambda T = 0.25$ remains. Equation (11) divides remaining process-plus-terminal cost by the training target scale. Equation (12) has a positive log-probability sign because it minimizes cost: above-baseline sampled cost reduces that action's probability under gradient descent. The advantage is stopped for actor differentiation; actor terms sum decisions within an episode and average episodes, while squared-value errors and entropy average within episodes before batch aggregation. Total loss adds 0.5 times value loss and subtracts entropy weight, linearly reduced from 0.01 to zero.

S3. Caps, acquired area, intervals and quality targets

A requested reporting cap queries the latest completed prediction at or below that cap on the single path planned for total B = 0.25. No policy is replanned for a smaller cap. Predictions are right-continuous step functions; the initial or final estimate is carried until the next event or total cap. MAE and RMSE pool physical specimens; Random averages absolute and squared losses within specimen before pooling. Process A and event-stage quantities average repeats within specimen, then specimens within source group, then the six source groups equally. These weights differ because development group sizes are 9,9,8,9,8,7.

Table S4. All five-cap results and actual acquired fractions (MAE in MPa). Full-precision RMSE and R^2 remain in same_cap_metrics.csv.

Method	Cap	MAE	Mean area	Min area	Max area
Center-first	0.0	58.551	0.000000	0.000000	0.000000
Center-first	0.0625	47.885	0.062350	0.061399	0.062407
Center-first	0.125	48.827	0.111012	0.109491	0.125000
Center-first	0.1875	49.094	0.172513	0.171900	0.187121
Center-first	0.25	48.027	0.234759	0.234676	0.235577
Geometry-spread	0.0	58.551	0.000000	0.000000	0.000000
Geometry-spread	0.0625	50.221	0.062018	0.061399	0.062130
Geometry-spread	0.125	46.189	0.124548	0.123164	0.124629
Geometry-spread	0.1875	46.555	0.187336	0.186034	0.187407
Geometry-spread	0.25	46.910	0.249970	0.249269	0.250000
Serpentine	0.0	58.551	0.000000	0.000000	0.000000
Serpentine	0.0625	49.731	0.062379	0.062130	0.062407
Serpentine	0.125	48.773	0.124592	0.124260	0.124629
Serpentine	0.1875	48.343	0.186805	0.186391	0.186851
Serpentine	0.25	48.013	0.249184	0.248521	0.249258
Random	0.0	58.551	0.000000	0.000000	0.000000
Random	0.0625	53.156	0.053603	0.046897	0.062500
Random	0.125	49.153	0.118768	0.109307	0.125000

Method	Cap	MAE	Mean area	Min area	Max area
Random	0.1875	48.108	0.186646	0.171598	0.187500
Random	0.25	46.886	0.246943	0.234492	0.250000
Learned-static	0.0	58.551	0.000000	0.000000	0.000000
Learned-static	0.0625	46.465	0.062401	0.062130	0.062500
Learned-static	0.125	48.064	0.124803	0.124260	0.125000
Learned-static	0.1875	46.784	0.187400	0.187130	0.187500
Learned-static	0.25	46.993	0.235297	0.234674	0.249991
Proposed	0.0	58.551	0.000000	0.000000	0.000000
Proposed	0.0625	43.324	0.057700	0.046897	0.062500
Proposed	0.125	43.164	0.119824	0.109120	0.124998
Proposed	0.1875	42.928	0.182555	0.171415	0.187407
Proposed	0.25	42.385	0.247174	0.234492	0.250000

Table S5. Saved paired differences, proposed minus control. Negative error differences favor proposed; these are existing 95% intervals, conditional on selection.

Control	Metric	Estimate	95% low	95% high	Improved n
Center	A	-4.878	-9.109	-0.764	29
Center	early_A	-3.050	-6.935	1.000	34
Center	MAE	-5.642	-10.912	-1.012	27
Geometry	A	-3.714	-7.283	0.337	33
Geometry	early_A	-3.109	-7.690	1.943	27
Geometry	MAE	-4.525	-8.276	-0.670	29
Serpentine	A	-5.314	-9.973	0.296	33
Serpentine	early_A	-3.399	-8.155	2.096	28
Serpentine	MAE	-5.628	-10.306	-0.477	34
Random	A	-5.997	-9.134	-2.601	35
Random	early_A	-5.346	-8.934	-1.375	33
Random	MAE	-4.502	-7.953	-0.950	34
Static	A	-4.380	-8.405	-0.199	25
Static	early_A	-4.175	-8.881	0.490	29
Static	MAE	-4.608	-8.628	-0.656	33
FULL_SCAN	MAE	0.695	-4.009	5.327	27

The percentile intervals use the existing 5,000 paired capture-group draws within source groups. Neither checkpoint selection nor multiple comparisons is adjusted. No interval is newly generated. The partial-minus-full MAE contrast of +0.695 MPa [−4.009, 5.327] is a difference estimate; no prespecified equivalence/noninferiority margin was tested. The area-cost axis measures supplied image pixels rather than measured scanner time.

Table S6a. Five-cap first-attainment caps for every preset target, including the full-input anchor. NR = not reached by 0.25; 0 = initially met; † = later recrossing. Relative saving is undefined if either method is NR or control cost is zero. All statuses and savings are preserved in matched_quality.csv.

Target	Center-first	Geometry-spread	Serpentine	Random	Learned-static	Proposed
41	NR	NR	NR	NR	NR	NR
42	NR	NR	NR	NR	NR	NR
43	NR	NR	NR	NR	NR	0.187500
44	NR	NR	NR	NR	NR	0.062500
45	NR	NR	NR	NR	NR	0.062500
46	NR	NR	NR	NR	NR	0.062500
47	NR	0.125000	NR	0.250000	0.062500†	0.062500
48	0.062500†	0.125000	NR	0.250000	0.062500†	0.062500
49	0.062500†	0.125000	0.125000	0.187500	0.062500	0.062500
50	0.062500	0.125000	0.062500	0.125000	0.062500	0.062500
51	0.062500	0.062500	0.062500	0.125000	0.062500	0.062500
52	0.062500	0.062500	0.062500	0.125000	0.062500	0.062500
53	0.062500	0.062500	0.062500	0.125000	0.062500	0.062500
54	0.062500	0.062500	0.062500	0.062500	0.062500	0.062500
55	0.062500	0.062500	0.062500	0.062500	0.062500	0.062500
56	0.062500	0.062500	0.062500	0.062500	0.062500	0.062500
57	0.062500	0.062500	0.062500	0.062500	0.062500	0.062500
58	0.062500	0.062500	0.062500	0.062500	0.062500	0.062500
59	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
60	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
61	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
41.690	NR	NR	NR	NR	NR	NR

Table S6b. Event-union first-attainment caps for every preset target, including the full-input anchor. NR = not reached by 0.25; 0 = initially met; † = later recrossing. Relative saving is undefined if either method is NR or control cost is zero. All statuses and savings are preserved in matched_quality.csv.

Target	Center-first	Geometry-spread	Serpentine	Random	Learned-static	Proposed
41	NR	NR	NR	NR	NR	NR
42	NR	NR	NR	NR	NR	0.078844†
43	NR	NR	NR	NR	NR	0.062591†
44	NR	NR	NR	NR	NR	0.062222
45	NR	NR	NR	NR	NR	0.046897†
46	NR	0.109120†	NR	NR	0.062407†	0.031203†
47	NR	0.093426	NR	0.249630	0.062407†	0.031203†
48	0.062407†	0.077732	NR	0.217830	0.062407†	0.031203
49	0.046897†	0.077732	0.109120†	0.125001	0.062130	0.015715
50	0.046897	0.062038†	0.062407	0.109304	0.046713	0.015715
51	0.046897	0.030700†	0.046713†	0.093610	0.046713	0.015715
52	0.031065	0.015509	0.015509†	0.077732†	0.046713	0.015694
53	0.015694	0.015509	0.015509	0.047337†	0.046713	0.015694
54	0.015694	0.015509	0.015509	0.031431	0.031203	0.015694
55	0.015694	0.015509	0.015509	0.031203	0.031203	0.015509
56	0.015694	0.015509	0.015509	0.031019	0.015902	0.015509
57	0.015350	0.015350	0.015350	0.015715	0.015715	0.015509

Target	Center-first	Geometry-spread	Serpentine	Random	Learned-static	Proposed
58	0.015350	0.015350	0.015350	0.015694	0.015694	0.015350
59	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
60	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
61	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
41.690	NR	NR	NR	NR	NR	NR

At q = 47 MPa, five-cap attainment is 0.0625 for proposed and learned-static, 0.125 for geometry and 0.25 for Random. The 50% reduction applies to geometry on this grid only. At q = 42 MPa the proposed event curve first crosses at 0.078844 and later recrosses; the five-cap grid is NR. The 41.690 MPa full-input anchor is not attained by 0.25. These are cohort first crossings, not an execution-time specimen-level stopping rule or engineering certification.

S4. Event contributions and fixed-set order interventions

Stage 1–4 completion intervals are (0,.0625], (.0625,.125], (.125,.1875] and (.1875,.25]. A raw decrease is previous absolute error minus updated absolute error. Its weighted contribution multiplies that decrease by (1 − completion fraction / B). Both retain adverse events. The sum of weighted decreases equals initial error minus process A. Archived methods use their original float64 area clock; fixed-set order intervention uses integer cumulative native pixels divided by total crop pixels. The two identity families remain separate; small native-replay differences from archived predictions are not silently mixed.

Table S7. All saved paired stage contrasts (MPa), proposed minus control; positive favors proposed. No direction change is applied to this already correct source table.

Control	Stage	Type	Gain	95% low	95% high
Center-first	1	weighted	4.176	-1.056	9.054
Center-first	1	raw	4.552	-1.246	10.032
Center-first	2	weighted	0.572	-2.116	3.205
Center-first	2	raw	0.727	-3.500	4.848
Center-first	3	weighted	0.161	-0.705	1.027
Center-first	3	raw	0.457	-1.753	2.608
Center-first	4	weighted	-0.031	-0.244	0.194
Center-first	4	raw	-0.590	-2.359	1.054
Geometry-spread	1	weighted	5.926	-0.117	11.095
Geometry-spread	1	raw	6.865	0.362	12.502
Geometry-spread	2	weighted	-2.475	-6.110	1.112
Geometry-spread	2	raw	-4.084	-9.387	1.182
Geometry-spread	3	weighted	0.199	-0.619	0.942
Geometry-spread	3	raw	0.635	-1.626	2.672
Geometry-spread	4	weighted	0.063	-0.083	0.217
Geometry-spread	4	raw	0.842	-0.597	2.243
Serpentine	1	weighted	5.562	-0.805	11.071
Serpentine	1	raw	6.282	-0.580	12.239
Serpentine	2	weighted	-0.340	-2.653	2.159
Serpentine	2	raw	-1.004	-4.757	2.916
Serpentine	3	weighted	-0.032	-0.886	0.830
Serpentine	3	raw	-0.051	-2.323	2.196
Serpentine	4	weighted	0.125	-0.046	0.305

Control	Stage	Type	Gain	95% low	95% high
Serpentine	4	raw	0.116	-1.293	1.664
Random	1	weighted	8.617	4.194	12.678
Random	1	raw	9.707	4.862	14.131
Random	2	weighted	-2.314	-4.549	0.070
Random	2	raw	-3.998	-7.409	-0.326
Random	3	weighted	-0.286	-0.954	0.347
Random	3	raw	-0.724	-2.547	1.021
Random	4	weighted	-0.019	-0.191	0.165
Random	4	raw	-0.713	-2.124	0.736
Learned-static	1	weighted	3.341	-1.651	8.050
Learned-static	1	raw	3.063	-2.381	8.149
Learned-static	2	weighted	1.254	-0.919	3.350
Learned-static	2	raw	1.597	-1.752	4.860
Learned-static	3	weighted	-0.365	-1.225	0.505
Learned-static	3	raw	-0.971	-3.312	1.405
Learned-static	4	weighted	0.150	-0.035	0.330
Learned-static	4	raw	0.694	-0.874	2.160

Table S8. Normalized fixed-set contrasts. N = native/original; C = comparator. Error rows are comparator minus native. Contribution rows are native minus comparator. All positive values favor native under these distinct definitions.

Contrast	Metric	Direction	Estimate	95% low	95% high
Primary permutation	A	C−N error	1.486	-0.658	3.384
Primary permutation	Early A	C−N error	1.628	-2.190	5.180
Primary permutation	J	C−N error	1.486	-0.658	3.384
Primary permutation	Final error	C−N error	0.000	0.000	0.000
Primary permutation	Weighted S1	N−C contribution	3.074	-0.578	6.502
Primary permutation	Weighted S2	N−C contribution	-1.255	-3.355	0.899
Primary permutation	Weighted S3	N−C contribution	-0.325	-0.940	0.250
Primary permutation	Weighted S4	N−C contribution	-0.008	-0.167	0.167
Secondary reverse	A	C−N error	3.929	0.145	7.405
Secondary reverse	Early A	C−N error	3.328	-1.527	7.979
Secondary reverse	J	C−N error	3.929	0.145	7.405
Secondary reverse	Final error	C−N error	0.000	0.000	0.000
Secondary reverse	Weighted S1	N−C contribution	6.348	0.695	11.784
Secondary reverse	Weighted S2	N−C contribution	-1.742	-4.553	1.118

Contrast	Metric	Direction	Estimate	95% low	95% high
Secondary reverse	Weighted S3	N−C contribution	-0.291	-1.385	0.844
Secondary reverse	Weighted S4	N−C contribution	-0.387	-0.744	-0.011

The legacy order_paired_contrasts.csv contains eight weighted-stage rows labeled comparator-minus-native. Only those contribution rows are negated in the derived normalized table, with CI [lo,hi] transformed to $[-\text{hi}, -\text{lo}]$. Eight error/objective rows remain unchanged. source_* columns preserve the original values and source Git blob 23a65f8d481797298b25bb228d1b2fd91181c65f. Applying normalization twice is identical to applying it once. The four normalized stage differences sum to the corresponding A difference within 10^{-8} MPa. Because final error is invariant within each fixed set, J − A differences also close. DIRECTION_CORRECTION.json records residuals.

The original-order intervention reproduced the archived predictions within the saved maximum deviation 6.104×10^{-5} MPa. All 350 terminal-set checks preserved the same terminal input and prediction. This is expected from a set-based predictor and validates the intervention; it is not an endpoint accuracy gain caused by order. The primary permutation A interval [−0.658,3.384] crosses zero, whereas the secondary reverse interval [0.145,7.405] is positive. Both early-A intervals cross zero. No component-specific causal claim or additive percentage decomposition follows.

S5. Six fixed cases and saved information-use diagnostics

The six cases were fixed before this revision and are all retained, including adverse residuals. Initial preferences are feasible-action softmax probabilities, not regional damage probabilities. The attribution maps are already saved L1 sensitivity of the selected action's log probability to local surface descriptors, including the path through the current strength prediction. Attention maps are the saved global-query-to-regional attention among fused tokens. These three quantities have different meanings and units; none is a regional mechanical ground truth. The displays below use only the initial, NO_VLM model's saved arrays. No attribution or attention was regenerated.

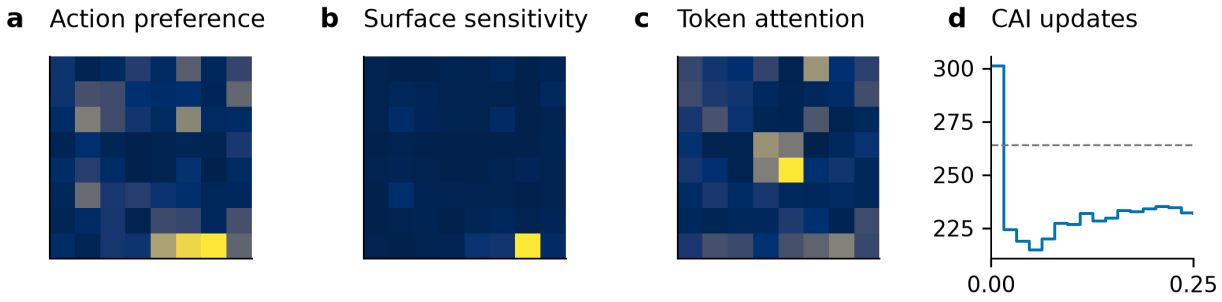
Table S9. All six cases; CAI values and errors in MPa.

Specimen	First cell	Actions	CAI target	Initial	Final	Final abs. error
74t7kcdgkr:c8-16	62	16	264.163	301.366	232.095	32.068
cgtnjyggtm:q24-48	31	16	136.680	302.099	258.932	122.252
w68dtmpfyf:q16-29	14	15	338.548	326.638	335.890	2.658
xcmzfsbd9t:c24-9t	27	16	278.254	225.969	233.148	45.105
yfxyg8jm46:c16-24	63	15	523.441	388.144	406.717	116.725
ykhs7s2dck:q8-22	63	16	235.013	245.533	244.583	9.570

Nine of 94 recorded decisions departed from the highest initially scored cell still feasible on the actual path: c8-16 (2), q16-29 (2), c16-24 (4), q8-22 (1); q24-48 and c24-9t retained their initial feasible ranking. The

comparison removes measured and unaffordable cells at every actual state. This describes specimen-specific initialization and selective reordering in six cases only. Changing internal evidence, history, progress and prediction are coupled, so the departures do not isolate internal-feedback causality.

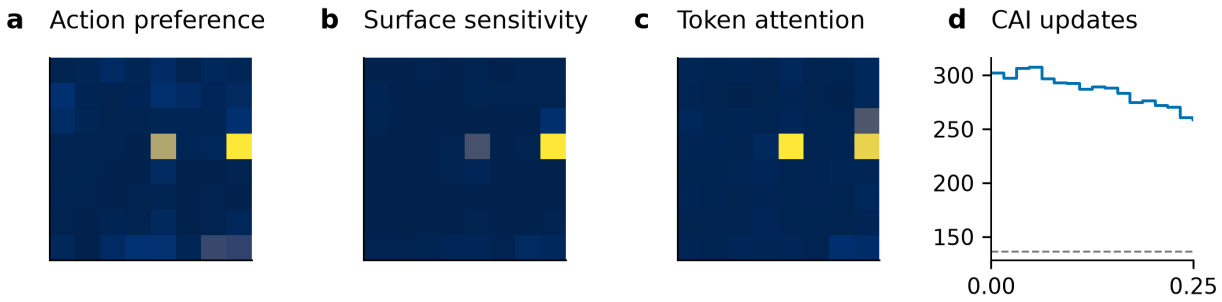
S5.1. 74t7kcdgkr:c8-16



74t7kcdgkr:c8-16 | Each map uses its own range; no cross-map quantitative scale.

Figure S5.1. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 62. Final absolute error: 32.068 MPa.

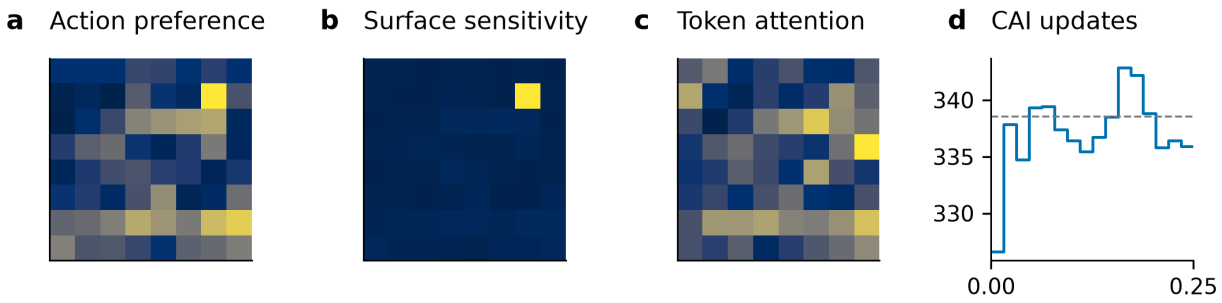
S5.2. cgtnjyggtm:q24-48



cgtnjyggtm:q24-48 | Each map uses its own range; no cross-map quantitative scale.

Figure S5.2. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 31. Final absolute error: 122.252 MPa.

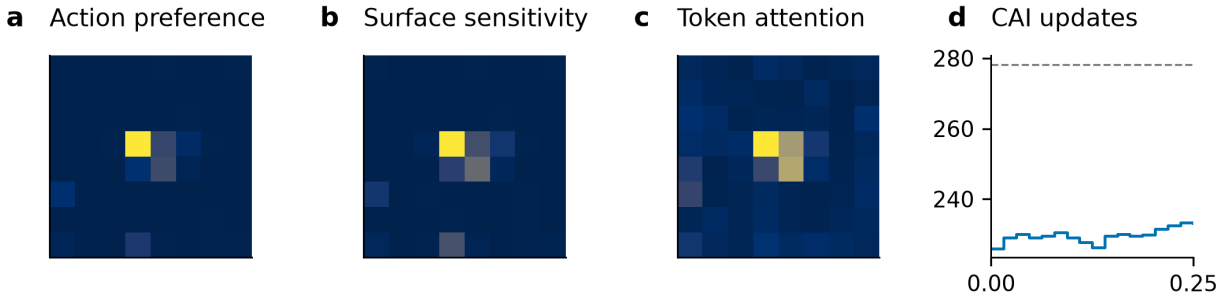
S5.3. w68dtmpfyf:q16-29



w68dtmpfyf:q16-29 | Each map uses its own range; no cross-map quantitative scale.

Figure S5.3. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 14. Final absolute error: 2.658 MPa.

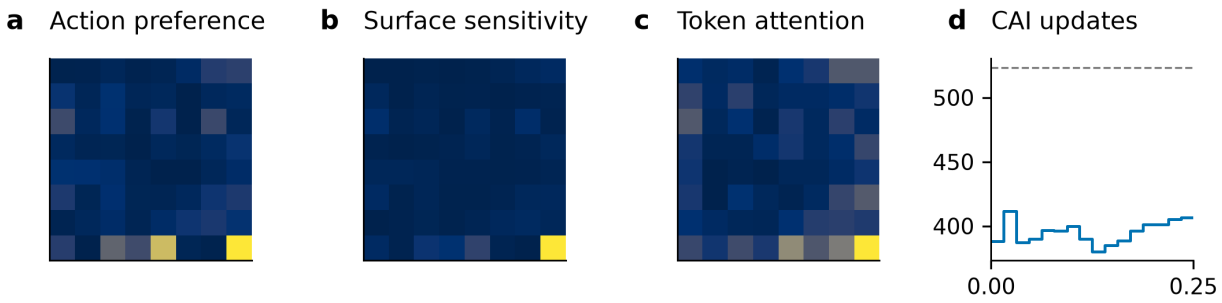
S5.4. xcmzfsbd9t:c24-9t



xcmzfsbd9t:c24-9t | Each map uses its own range; no cross-map quantitative scale.

Figure S5.4. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 27. Final absolute error: 45.105 MPa.

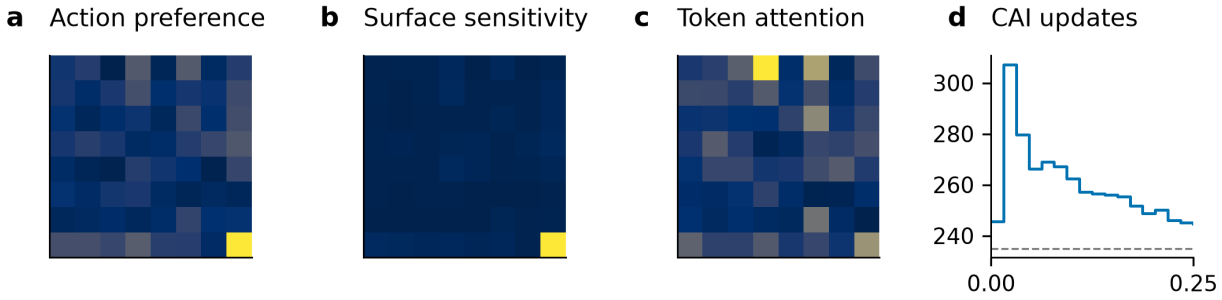
S5.5. yfxyg8jm46:c16-24



yfxyg8jm46:c16-24 | Each map uses its own range; no cross-map quantitative scale.

Figure S5.5. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 63. Final absolute error: 116.725 MPa.

S5.6. ykhs7s2dck:q8-22



ykhs7s2dck:q8-22 | Each map uses its own range; no cross-map quantitative scale.

Figure S5.6. Initial NO_VLM state and saved CAI trajectory. (a) Feasible-action preference; (b) surface L1 sensitivity for the selected-action query (first saved target); (c) first-layer global-query regional attention (first saved layer); (d) saved CAI predictions with measured strength dashed. Maps retain all 64 cells and use independent ranges. Values and target/layer indices are in figure_data/cases.json; selected cell 63. Final absolute error: 9.570 MPa.

S5.7. Real-image provenance and reproducible displays

Main Fig. 6 uses the source-qualified c8-16 and q24-48 image pairs from Mendeley Data version 1, collections 74t7kcdgkr and cgtnjyggtm (Hasebe, Higuchi, Yokozeki and Takeda, 2022; CC BY 4.0). The candidate manifest resolves each surface and registered C-scan path with SHA256 and native dimensions. Surface orientation follows the frozen clockwise-90° transform and maximum-edge-1024 display resize. The registered C-scan pixels are unchanged; overlays use rint native grid edges and the saved action ordinals. Grey overlays mark unavailable cells. The full-image backdrop is explicitly retrospective and was never supplied to the actor. assets.json records source URLs, hashes, dimensions, transformations and attribution. No damage outline, scale calibration or physical length measurement is invented.

The delivered scripts bind, prepare, render, build and verify this frozen-evidence package. Machine-readable trajectory ledgers provide all 500 archived paths and all 350 fixed-set conditions, without duplicating weights or feature shards. The eight document outputs provide English and aligned bilingual main text and supplement. Reference numbering is inherited from the main article; this supplement introduces no new references. Author names, affiliations, funding, competing interests and final journal submission metadata require author confirmation before submission; the package is a revision delivery, not a journal submission.